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The recovery of potable water from wastewater is critical for life support and the environmental health of crew members during long term space missions. The wastewater stream is contaminated with particulates, microorganisms, detergents, salts, and organic compounds, all of which must be removed before the water is suitable for reuse. Standard terrestrial techniques for water treatment and recovery are not suitable for aboard spacecraft. However, rotating reverse osmosis (RO) systems, the focus of this research, show promise for water treatment and recovery.
Rotating membrane systems in which a cylindrical filter rotates in a cylindrical shell setting up internal fluid flows have potential to greatly reduce the rate at which the filter loads with membrane foulants. This can substantially extend the life of water filtration systems requiring far less frequent change out of expendable components than standard filtration. Furthermore, because of their anti-fouling characteristics rotating membrane systems are applicable to ultrafiltration (UF) and RO to generate pure water. In our previous NASA-funded research, we investigated the potential for

Task Description:

combining rotating membrane filtration with RO to remove dissolved chemical species. Our initial measurements of rejection and flux for rotating RO membranes and our theoretical model indicate that rotating RO has higher flux and rejection than standard RO systems while simultaneously achieving a greater recovery.

Our previous work established the concept of a low-pressure rotating RO membrane system. A mathematical model based on the film theory and the solution-diffusion model agreed well with the experimental results obtained using this first generation prototype (Lee and Lueptow 2002). Based on the model, the filtrate flux and contaminant rejection depend more strongly on the transmembrane pressure than on the rotational speed (Lee and Lueptow 2001). The first generation apparatus was built to provide proof of the concept for rotating RO and was not capable of operating at high pressures or for long time periods. Therefore, the goal of our current work has been to improve the flux of the device by increasing the transmembrane pressure up to 500 psi and enabling long term testing (7 to 60 days) of the device. To accomplish this, a second generation prototype of a rotating reverse osmosis filter and the accompanying flow circuit was designed and fabricated. The solution diffusion thin film modeling of the system was also continued. Due to the need for long-term testing with high pressures, the operational mode of the second generation apparatus was altered. The theoretical model was used to determine the optimal design geometry of the second generation prototype.

To further enhance the second generation prototype, the performance of commercially available reverse osmosis membranes was evaluated using a laboratory scale batch filtration method. The permeate flux and rejection of ionic solutes and dissolved organic carbon were measured experimentally to select best performing membranes among seven types of membranes (five RO and two nanofiltration) with three different ersatz wastewater solutions simulating transit mission wastewater, early planetary base wastewater, and biological water process effluent. From these screening tests, two RO membranes having both high rejection and flux characteristics are being investigated for a rotating reverse osmosis system in space applications.

A separate supplemental study was conducted using ion exchange textiles as organic supports for urease immobilization with the aim of developing reactive fibrous materials able to promote urea removal. We have shown that urease can be successfully immobilized on cation exchange textiles using various coupling agents. The urease immobilized onto cation exchange textile offers various advantages. First, the textile (nonwoven fabric) form offers high surface-to-volume ratio and packing density. Furthermore, ion exchange resins in the form of beads move around resulting in abrasion which produces fine particles and degrades the surface. Using an ion exchange textile can prevent such abrasion maintaining the active surface area and preventing plugging downstream. Second, the immobilization procedure used here is simple to carry out without the need for complicated immobilization methods or the use of electron beam or gamma irradiation. Third, the activation energy of the urease immobilized is nearly as low as that of free urease, suggesting that the enzymatic reaction of the immobilized urease is nearly as good as that of the free urease. Fourth, although the kinetic parameters are slightly less favorable than those of free urease, the good storage stability and operational stability of the immobilized enzyme suggest superior potential of the immobilized enzyme for practical application. Finally, a preliminary study of the performance of the biotextile in a continuous flow reactor indicates that the biotextile can convert over 80 % of the urea over a period of several days.

Human exploration and utilization of space requires the development of a Water Recovery and Management System (WRMS) that minimizes the need for the external resupply of water. This is because space missions of longer duration, such as a mission to Mars, or having larger crews cannot depend on water storage or resupply. The WRMS must effectively purify the wastewater to generate an ongoing supply of potable water. The potable water produced from the wastewater must meet stringent specifications for purity. Trace levels of organic and inorganic contaminants are particularly of concern because of the potential for some compounds to accumulate in human body tissue. Microorganisms that may be pathogenic or clog water lines must be removed. In addition, flavor and clarity are important issues. Design requirements of a WRMS are also very stringent (Jan et al. 1996). The WRMS should be inherently reliable, capable, and efficient. The use of expendables should be minimal. Minimizing the total weight, volume, power consumption, and cost of the system while ensuring safe operation is necessary. Mass loop closure, in which all water is reclaimed, is essential for long-term missions. Therefore, the development of WRMS for long-term space missions is challenging.

The ultimate goal of our current research is to bring the rotating membrane technology closer to practice for recycling water in manned space missions. Rotating reverse osmosis (RO) is a promising technology for wastewater processing. RO has long been in use as a physical membrane separation technology. However, concentration polarization decreases the flux of solvent through the membrane and the rejection of contaminants as a result of an increase in the solute concentration near the membrane surface (Porter 1972). Rotating filtration in which a cylindrical filter rotates within an outer cylindrical shell has been shown to exhibit improved filtration (Hallstrom and Lopez-Leiva. 1978; Lee and Lueptow 2001). This has been attributed to the disruption of the concentration polarization layer by the high shear at the membrane surface due to the rotation and the presence of Taylor vortices in the annulus of a rotating device (Schwille et al. 2002). Therefore, rotating RO, which combines rotating filtration with RO, has a high selectivity typical of reverse osmosis along with improved flux and rejection due to the rotation. These improvements allow for operation of rotating RO at much higher recoveries than conventional RO systems.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

We have demonstrated that rotating RO technology has potential for producing high quality drinking water while maintaining a high recovery. Because of the enhanced mass transfer induced by vortical flow, rotating RO shows lower concentration polarization, resulting in high flux and high rejection. We have found that rotating RO can improve flux and rejection with a negligible energy cost due to the necessity to rotate the inner cylinder. We have designed and tested an improved rotating RO membrane module through both theoretical and experimental approaches to start optimizing the design of a rotating RO module. Particularly, we designed a module to operate at high pressures for long term tests. In addition, over the past year we have developed a urease biotextile to decompose urea to ammonium carbonate in space mission, which is easily removed by RO.
Our accomplishments over the last two years are shown below:

• Prediction of the performance of rotating RO for a variety of operating conditions

Task Progress:	• Measurement of the effectiveness of rotating RO for a variety of operating conditions • Experimental verification of our theoretical model for rotating RO • Process modeling and simulation for design and operation of a water reuse system • Evaluation of rotating RO under conditions for calcium and silica scale formation • Demonstration of anti-scaling effects in rotating RO • Detection of colloidal silica suspension concentration using fluorescence spectroscopy • Characterization of RO and NF membranes in terms of flux and rejection with ersatz wastewater solutions • Design and fabrication of a second generation rotating RO system for high pressure and long term operation • Urea hydrolysis using urease immobilization on an ion exchange textile.
Bibliography Type:	Description: (Last Updated: 06/28/2021)
Articles in Peer-reviewed Journals	Lee S, Lueptow RM. "Model prediction and experiments for rotating reverse osmosis for space mission water reuse." Separation Science and Technology. 2005;39(3):539-61. https://doi.org/10.1081/SS-120027994 , Mar-2005
Articles in Peer-reviewed Journals	Lee S, Lueptow RM. "Rotating membrane filtration and rotating reverse osmosis." Journal of Chemical Engineering of Japan. 2004 Jul;37(4):471-82. https://doi.org/10.1252/jcej.37.471 , Jul-2004
Articles in Peer-reviewed Journals	Lee S, Lueptow RM. "Rotating reverse osmosis for water recovery in space: influence of operational parameters on RO performance." Desalination. 2004 Oct 1;169(2):109-20. https://doi.org/10.1016/j.desal.2004.08.012 , Oct-2004